

Design of Asphalt Mixes

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Terminology and Mix Designation



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Types of Mixes

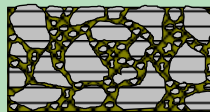
- AC (Asphaltic Concrete) = Dense Graded Hot Mix
- OGPA Open Graded Porous Asphalt
- SMA Stone Mastic Asphalt

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Types of Asphalt Mixes



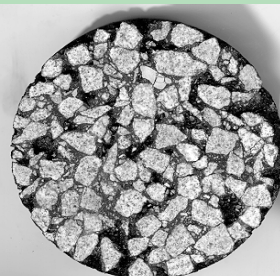
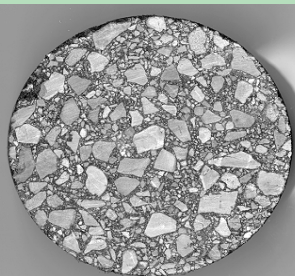
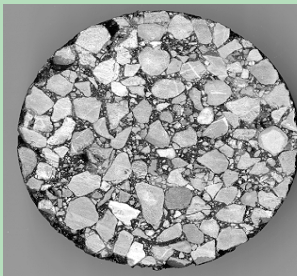
SMA



Dense Graded Mix



OGPA



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Terminology

- Nominal Size: a convenient whole number greater than largest sieve size to retain more than 0% and less than 10% of the aggregate material
- Compaction Levels: Three levels of compactions: 50, 80 and 120 cycles based on the expected traffic level.
- Binder Type or Class:
 - Australian Standard Grades 320, 450 & 600
 - New Zealand PG 64, 58, 54 – S,H,V,E (M01A)
- Additional Suffix: This may be used to special mix requirements for example R for bitumen rich mix

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Special mix type to meet particular requirements

Mix Type	Use
Open Graded	Noise or spray reduction
Stone mastic asphalt	Coarse textured mix with good rutting resistance
Fine gap graded	Durable mix for low traffic situations

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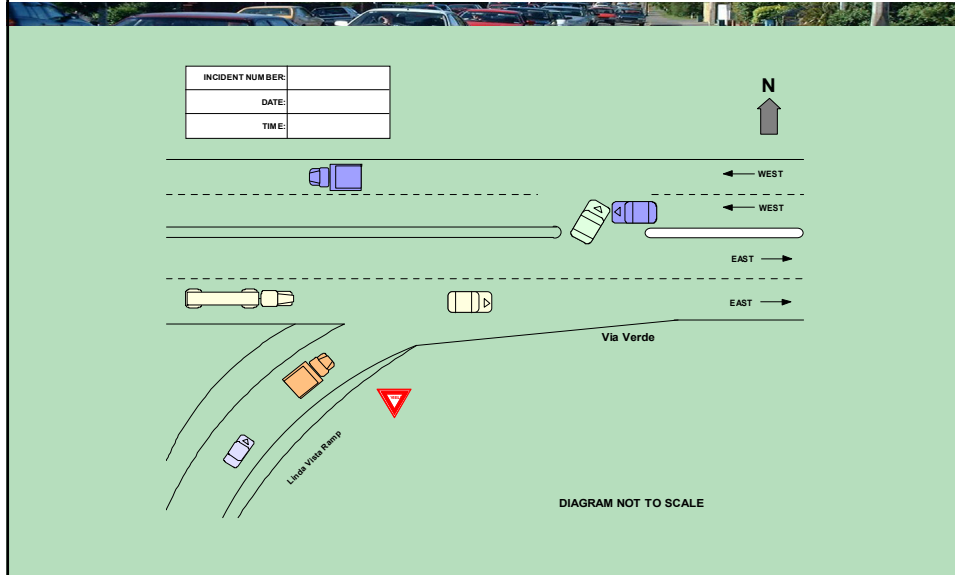
Factors to be Considered in Selecting Mix Type

- Structural Requirements
 - The pavement configuration may impose high fatigue resistance requirement.
 - The pavement design may call for minimum modulus values.
- Layer Thickness
 - interacts with the structural properties to determine pavement deflection under load and thus fatigue requirements of the mix

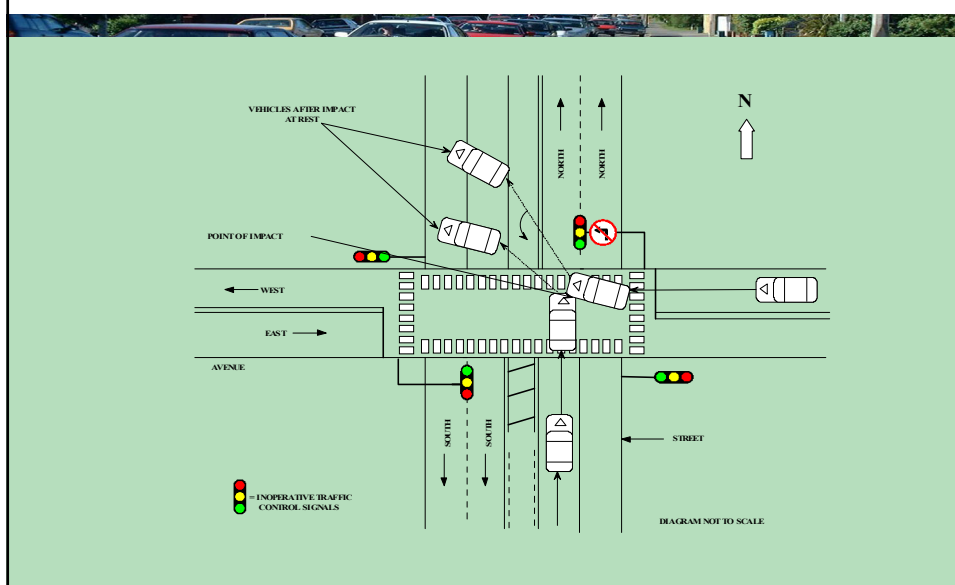
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- Traffic Type, Volume and Speed
 - High traffic volume and low speed impose the most severe rutting conditions.
 - Rutting is mainly caused by heavy commercial vehicles and exacerbated by super single tyres.
- Pavement Geometry
 - High shearing forces may be generated at the surface by turning, braking and accelerating traffic, and at stop situations on inclines.

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- Climate Conditions
 - Hot weather promotes rutting, and low temperature promote fatigue and thermal fatigue cracking
- Surface Requirements
 - Good skid resistance may require polish resistant aggregate, and good surface texture where traffic speeds are high.
 - Noise and spray can be substantially reduced through the selection of the appropriate mix type.
- Material and Equipment Available
- Cost

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Performance Requirements

Residential Streets Mixes

Low %Va
(High Density)
Impermeable yet
still Flexible

Finer grading
Gap type grading
High binder content
Smooth rounded Fines
Softer grade of binder

Motorways/Freeways Mixes

High Stability to resist rutting
Binder content sufficient for
Cohesion
Fatigue resistance

Coarse Grading
Angular/crushed fines
Rough textured aggregate
Hard Grade of binder
Modified binder

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Traffic Classification (NZTA M10:2020)

HCV/Lane/ Day	ESA	Traffic Category	
		Free flowing	Stop/start or climbing or slow moving
<100	<0.5 *10 ⁶	Light	Medium
100 to 500	0.5 *10 ⁶ to 5*10 ⁶	Medium	Heavy
500 to 1000	5 *10 ⁶ to 20*10 ⁶	Heavy	Very Heavy
>1000	>20 *10 ⁶	Very Heavy	Very Heavy

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Guide to selection of Dense Graded Mix parameters

Traffic Category	Lab Compaction Level	Design Air voids	Binder Class (M1A)	Recommended Use
Light	50 Cycles	4.0	S	Residential Streets
Medium	80 Cycles	4.0	H	Intersections & Collectors
Heavy	120 Cycles	4.0	V	Heavy trafficked applications; motorways
Very Heavy	120 and 250 Cycles	4.0%	E	Slow moving, heavily trafficked roads; ports

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APRG18 Mix Design

- APRG18 dense graded design procedure was developed on performance based principles by a committee of experts from industry and public sector
- Incorporates outputs of a coordinated laboratory research program, supported by an Accelerated loading Facility (ALF) trial on full scale pavements. Some elements of US superpave design procedure have been incorporated into the APRG18.
- APRG18 is basis of NZTA M10:2020 Specification for Dense Graded Asphaltic Concrete

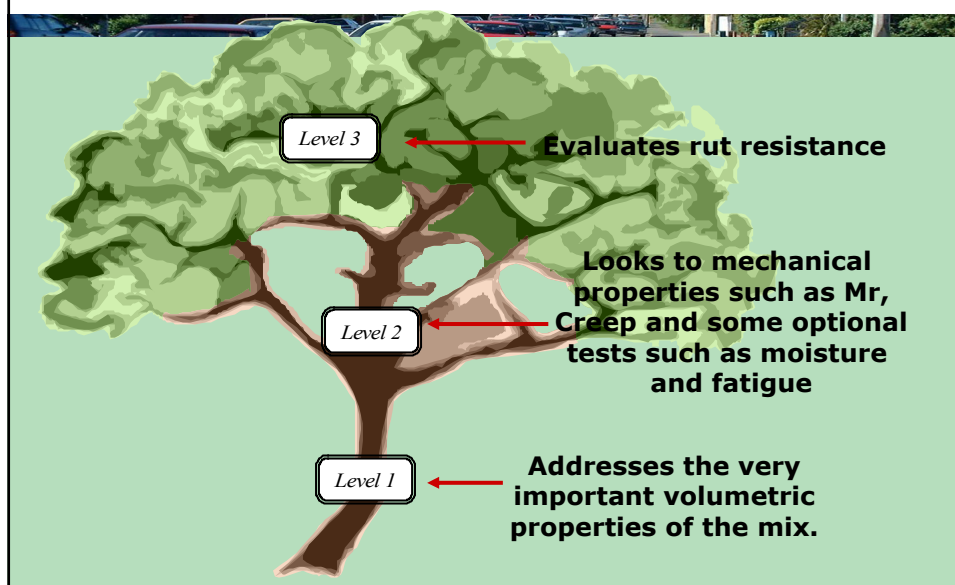
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Performance Requirements

- The APRG18/M10 procedure is based on four key performance requirements:
 - Permanent deformation (surface courses)
 - Fatigue
 - Moisture Susceptibility
 - Durability

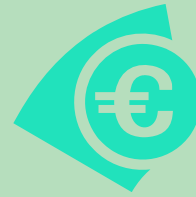
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APRG18 Design Levels



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- Cost and resources required for mix design testing must be balanced against the circumstances in which the mix will be used



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Design of Dense Graded Mixes by Gyratory Compactor

Only small amount of testing is required for low traffic mixes and they are only required to undergo level 1.

- Medium and heavy traffic mixes undergo level 1 and level 2 testing.
- For very heavy traffic volumes, level 1, level 2 and level 3 testing are essential.

Performance tests, such as wheel tracking and beam fatigue, are expensive. One of the aims of level 1 stage of the procedure is to select a composition which is likely to satisfy the wheel tracking and fatigue requirements before these properties are tested in the level 2 and level 3.

Another very important aim of level 1 is to ensure satisfactory performance of those mixes not required to undergo level 2 and level 3 testing.

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Summary of Level 1

- This level deals with volumetric testing and is intended for light traffic situations.
- Compaction level is 50 gyrations
- Four or five binder contents are chosen for new mix.
- Volumetric properties versus binder content charts are constructed.
- Design binder content is the one that yields 4.0% air voids.

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Summary of Level 2

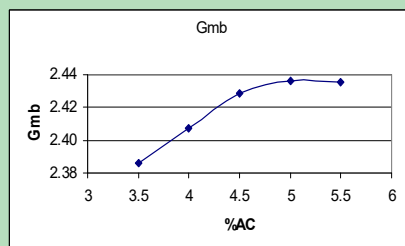
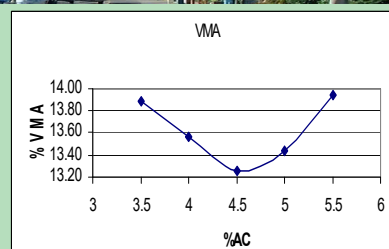
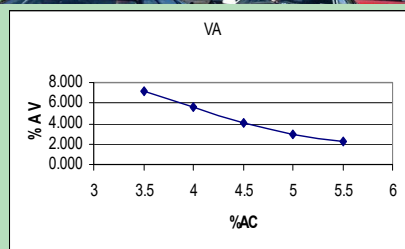
- This level deals with volumetric testing, in addition, mechanical properties are intended for medium and heavy traffic situations.
- For Medium Traffic, compaction level is 80 gyrations
- For Heavy traffic, compaction level is 120 gyrations
- Four or five binder contents are chosen for new mix.
- Volumetric properties versus binder content charts are constructed.
- Design binder content is one that yields 4.0% air void.
- Mixes at the design binder content undergo resilient modulus and creep tests. Fatigue and moisture resistance tests are optional.

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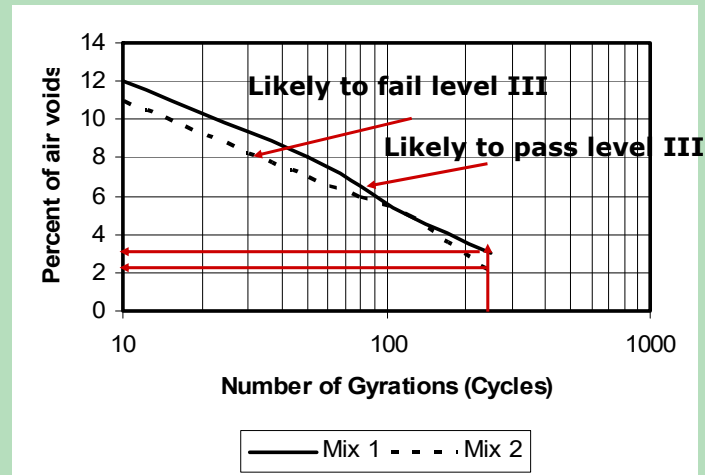
Summary of Level 3

- This level is intended for very heavy traffic situations.
- Compaction level is 120 gyrations
- Four or five binder contents are chosen for new mix.
- Volumetric properties versus binder content charts are constructed.
- Design binder content is what gives **4.0% air voids in NZTA M10**
- Specimens at the design binder content are subjected to a range of compaction levels from 10 to 250 cycles. The minimum air voids at 250 is 2.5%. The voids at 10 cycles should also be recorded.

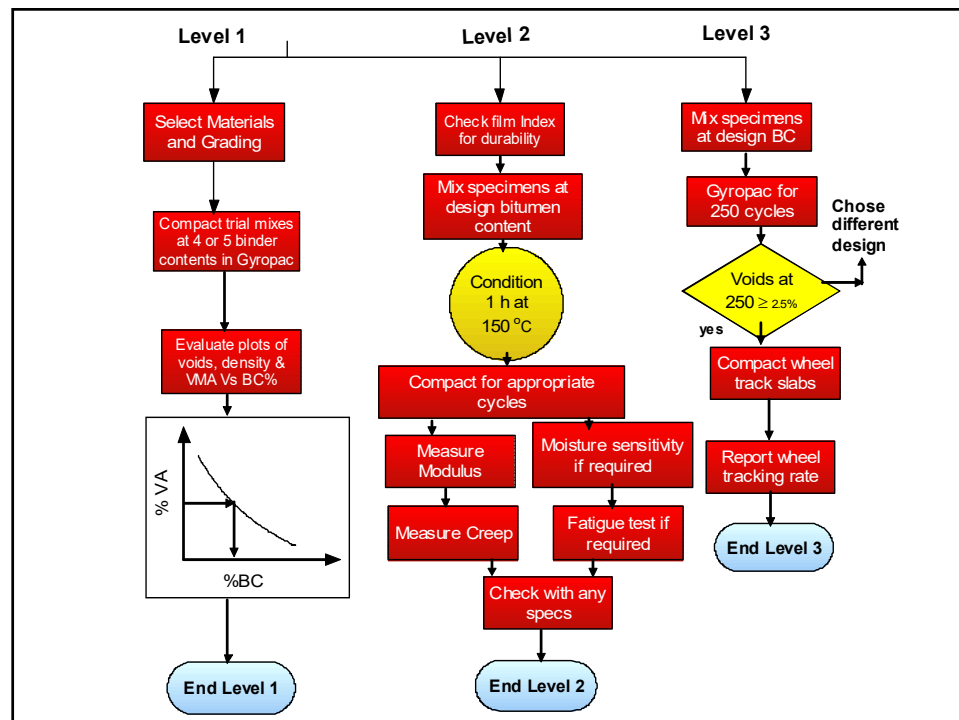
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Simulation of field conditions

- Important aim of procedure is to ensure that specimens prepared in laboratory have properties similar to asphalt placed on the road
- Design procedure therefore includes a laboratory conditioning step that simulates the binder hardening which occurs during manufacture and placement of mix, and the first year or two of field service

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Level 1 Testing

Selection of Materials, aggregate Grading and Trial Binder Content

- **Aggregate Selection and gradation**
 - *Aggregate grading and aggregate components are selected to suit the mix type being designed and the expected traffic conditions (See Table 1).*
 - *Aggregate properties such as size, shape, surface texture, cleanness and toughness have a significant effect on the final mix. Table 2 shows different nominal sizes of aggregate as a function of the layer thickness and the recommended use.*

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Grading Envelopes (M10)

		AC10	AC14	AC20	AC28
Sieve Size (mm)	Percent Passing sieve size by mass				
37.5					100
26.5			100	90-100	
19		100	90-100	73-88	
13.2	100	90-100	71-86	58-76	
9.5	90-100	72-83	58-75	47-67	
6.7	68-82	54-71	46-64	37-58	
4.75	50-70	43-61	37-55	30-50	
2.36	32-51	28-45	24-42	20-37	
1.18	22-40	19-35	15-32	13-28	
0.6	15-30	13-27	10-24	9-22	
0.3	10-22	9-20	7-17	6-16	
0.15	6-14	6-13	4-12	4-10	
0.075	4-7	4-7	3-6	3-6	

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Nominal Size	Typical layer Thickness (mm)	Typical Use
5	15-20	For use a very thin surfacing layer with fine surface texture
7	20-25	Surfacing residential streets
10	40-60	General purpose wearing course mix suitable for light and moderate traffic applications
14	55-80	Wearing course mix for heavy traffic application
20	80-120	General purpose base and intermediate course mix
28	100-160	Less commonly used for base. Segregation is a concern.

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- To produce a durable mix, it is necessary to ensure that the binder film thickness is adequate and a minimum film index value is commonly specified to achieve this.
- A trial binder content is chosen, either on the basis of past experience with similar mixes, or so that a minimum binder film index of 7.5μm or greater is obtained. This ensures the mix has adequate durability and fatigue resistance.
- Trial binder content = Min. BC% +0.5%

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Selecting Target Bitumen Content

$$F = \frac{P_{be}}{100 - P_b} * \frac{1}{C} * \frac{1000}{\rho_b}$$

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- F = Film thickness (μm)
- P_{be} = Percentage of effective binder content by mass of the total mix
- P_b = percentage of the total binder by total mix
- C = Surface area of aggregate (m²/kg)
- ρ_b = density of binder at 25 °C (t/m³)

$$P_{be} = P_b - \frac{P_{ba}}{100} * P_s$$

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$$P_{ba} \% = \frac{G_{se} - G_{sb}}{G_{se} * G_{sb}} * G_b * 100$$

P_{ba} is the percentage of the absorbed bitumen by weight of dry aggregates.

G_{se} = is the effective specific gravity

G_{sb} = bulk specific gravity

The surface area can be calculated as follows

$$C = (2 + 0.02 * P_{\#4} + 0.04 * P_{2.36} + 0.08 * P_{1.18} + 0.14 * P_{0.6} + 0.3 * P_{0.3} + 0.6 * P_{0.15} + 1.6 * P_{\#200}) * 0.20482$$

$$\text{If } G_{sb} = 2.65 \pm 10\% * 2.65 \quad 2.385 \leq G_{sb} \leq 2.915$$

If G_{sb} out of the above range use C*

$$C^* = C * 2.65 / G_{sb}$$

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Example

- If the effective specific gravity of aggregate is 2.8 and the bulk specific gravity is 2.7 and the aggregate gradation coincide with AC20. The bitumen specific gravity can be assumed 1.02. Determine the minimum bitumen content that provides the minimum thickness (7.5 μm)

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Assume $P_b = 4.0\%$ $P_s = 96.0\%$

$$P_{ba} = \frac{2.8 - 2.7}{2.8 * 2.7} * 1.02 * 100 = 1.35\%$$

$$A = (2 + 0.02 * 50 + 0.04 * 34 + 0.08 * 27 + 0.14 * 21 + 0.3 * 15 + 0.6 * 11 + 1.6 * 5.5) * 0.20482 = 6.01$$

$$A = 6.01 \text{ kg/m}^2$$

$$P_{be} = 4.0 - \frac{1.35}{100} * 96 = 2.704$$

$$F = \frac{P_{be}}{100 - P_b} * \frac{1}{A} * \frac{1000}{\rho_b} = \frac{2.704}{100 - 4.0} * \frac{1}{6.01} * \frac{1000}{1.02} = 4.59 \mu\text{m} < 7.5 \mu\text{m}$$

Assume $P_b = 5.0$ then $P_s = 95\%$

$$P_{be} = 5.0 - \frac{1.35}{100} * 95 = 3.72 \%$$

$$F = \frac{3.72}{100 - 5.0} * \frac{1}{6.01} * \frac{1000}{1.02} = 6.38 < 7.5 \mu\text{m}$$

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Make a third trial assume $P_b=5.5\%$ then $P_s=94.5\%$

$P_{be}=4.22$

$$F = \frac{4.22}{100 - 5.5} * \frac{1}{6.01} * \frac{1000}{1.02} = 7.3 \approx 7.5 \mu m$$

Minimum bitumen content (MBC%) = 5.5%

Trial bitumen content (TBC%) = $5.5 + 0.5 = 6.0\%$

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Mixing, Conditioning, and Compaction

- Usually **PG64 or 58 V or E** pen grade bitumen is used for asphalt work in medium & heavily trafficked areas and should be used unless otherwise specified. To conduct the volumetric design of level 1, four to five bitumen contents will be used to produce three replicates at each bitumen content in addition to secure enough amount of loose mix to carry out the maximum specific gravity test. For redesigns or for materials with previous experience four bitumen contents may be used.
- These bitumen contents are normally the trial bitumen content (TBC) & $TBC \pm 1.0\%$. Depending on the results of testing, 1 or 2 mixes with 0.5% difference from those already prepared would be used to fill the gaps or extend the binder content range in the appropriate direction.

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Gyratory compactors



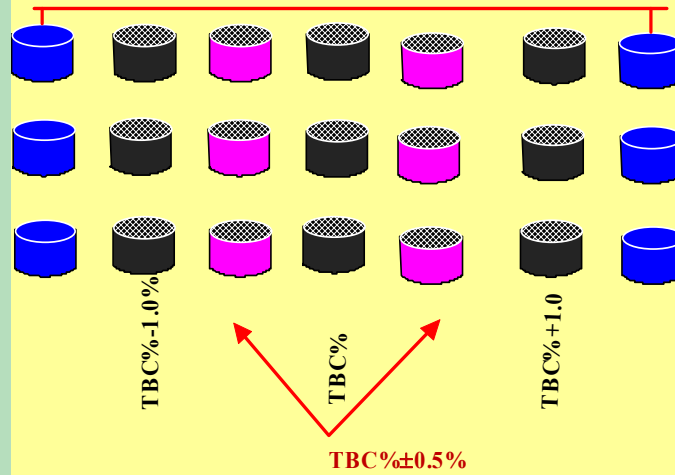
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- For each batch, the selected binder and aggregate is heated in an oven and mixed at about 150 °C using manual or mechanical mixing. The mix is conditioned at 150 C for an hour. This conditioning simulates the hardening of the asphalt mix occurs in the mixing plant and during transport and lay down, compaction, and the first year or two of service.
- Three replicate samples are mixed at each bitumen content and conditioned as described. The samples are then compacted for the required number of cycles using gyratory compactor.
- For mixes in the very heavy traffic category only, following determination of the design binder content, a further set of samples are prepared. These are made at the design binder content and duplicate samples are compacted at 10, 80, 120, and 250 cycles. Voids at 250 cycles are then checked.

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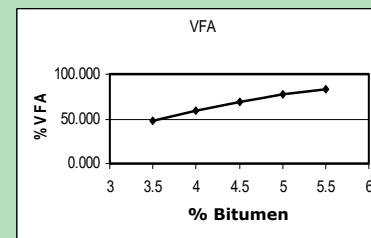
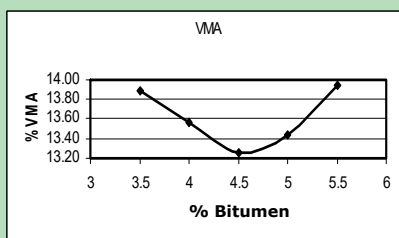
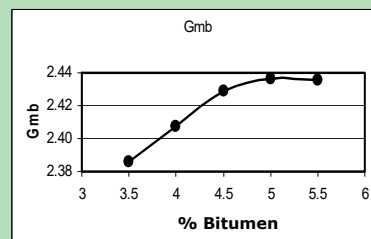
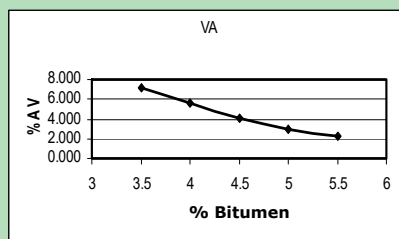
Compaction Level 50 or 80 or 120

OR TBC $\pm$ 1.5%



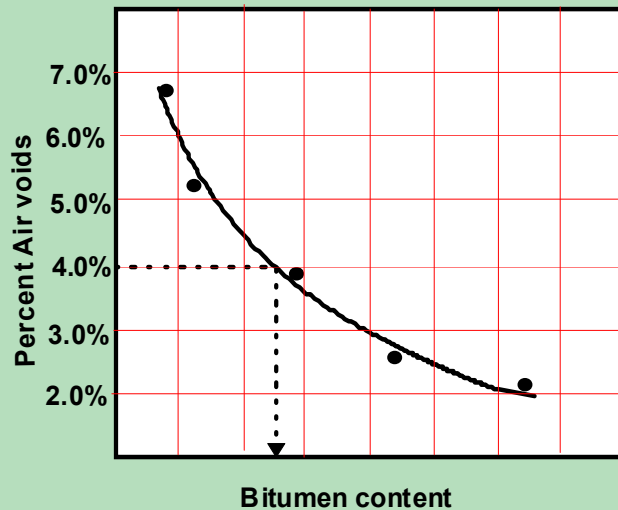
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Step # 3: Determination of the Design Binder Content



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Determining bitumen content



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Level II Testing

- Since level two represents mixes designed for moderate and heavily trafficked areas, the cost of failure of such roads will be dramatically much higher than that could be occurred on lightly trafficked areas. Also, it is not acceptable by road users to have such failures on these roads. For these reasons, it is appropriate to invest more time and effort in the design of moderately and heavily trafficked roads. In level II, the mechanical testing on the trial asphalt concrete mix will be carried out.
- Where a PMB is to be used, the purchaser may request that the modulus and creep testing be first carried out using class 320 bitumen to ensure that a satisfactory aggregate structure is obtained. A further set with PMB is then manufactured and tested for compliance with specification requirements.

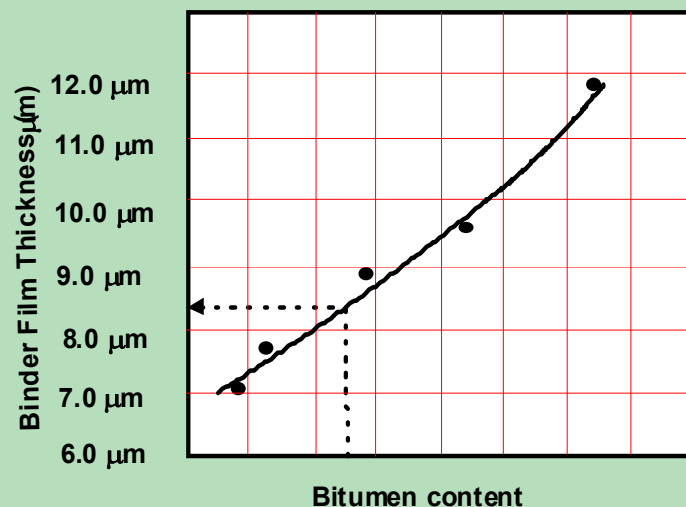
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Step # 4: Bitumen Film Thickness Check

- Check that a minimum 7.5 μm cover is achieved. This minimum thickness will ensure cohesion, durability and fatigue resistance of the design mix. An alternative way to check the bitumen thickness is to check the percentage of air voids filled with binder and make sure that it satisfies the design criteria.

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Check Film Index or %VFB



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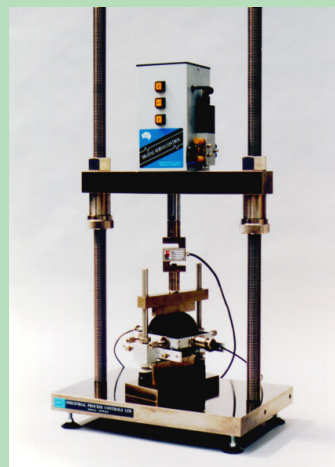
Step # 5: Prepare Specimens for Stiffness, Creep and Moisture Sensitivity

- A triplicate set of specimens at the design bitumen content is prepared at a percentage air void 5.0% using the same process of mixing and conditioning followed in level 1 design. After preparing the specimens, check the bulk density and calculate the percent air voids and make sure that it is within $4.0 \pm 0.5\%$.
- Also prepare another two set of specimens each set is three specimens for measuring the moisture sensitivity. Specimens for moisture sensitivity are prepared at percentage of air voids 8.0 ± 1.0 which represents the critical air voids for stripping. Try to make sure that the average air voids of the two sets is almost the same.

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Step # 6: Measurements of Modulus

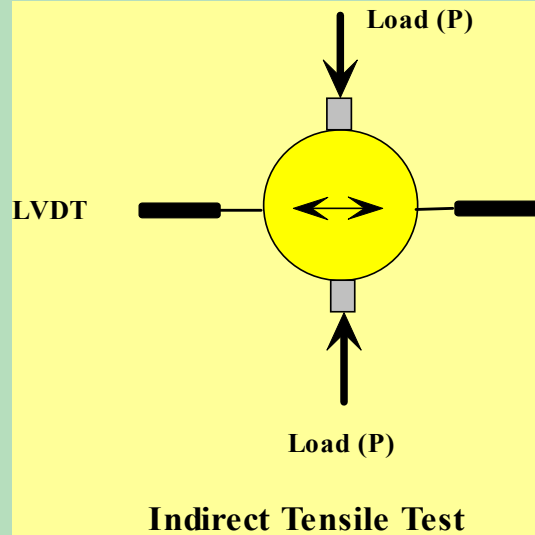
- Indirect tensile test is carried out at 25 °C on a set of the triplicate specimens & an average value of modulus is determined
- Check that average modulus meets design requirements; if not, change design mix and repeat previous steps



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Indirect Tensile Test

$$M_r = \frac{P * (0.273 + \nu)}{\Delta * h}$$

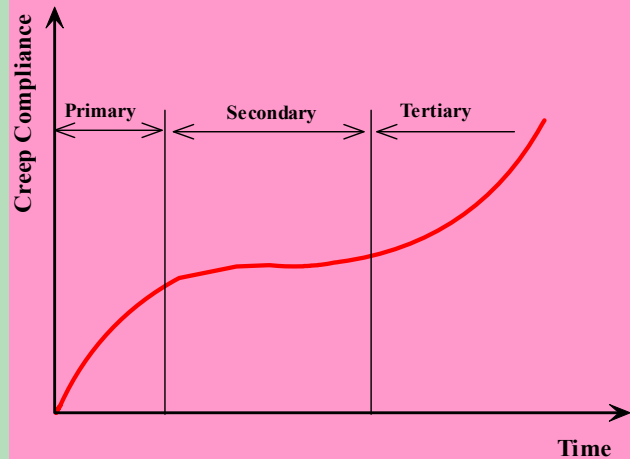


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Step #7: Measurement of Creep

- Creep test is used to determine the resistance to deformation of the asphalt mix
- Dynamic creep test is performed on same set of specimens used in indirect test but applying axial loading at 50 °C
- Mean minimum slope is derived from the dynamic creep data and compared to the minimum requirements (based on climate, traffic level and traffic speed). If the results do not comply with specification, mix composition is readjusted and process is repeated again

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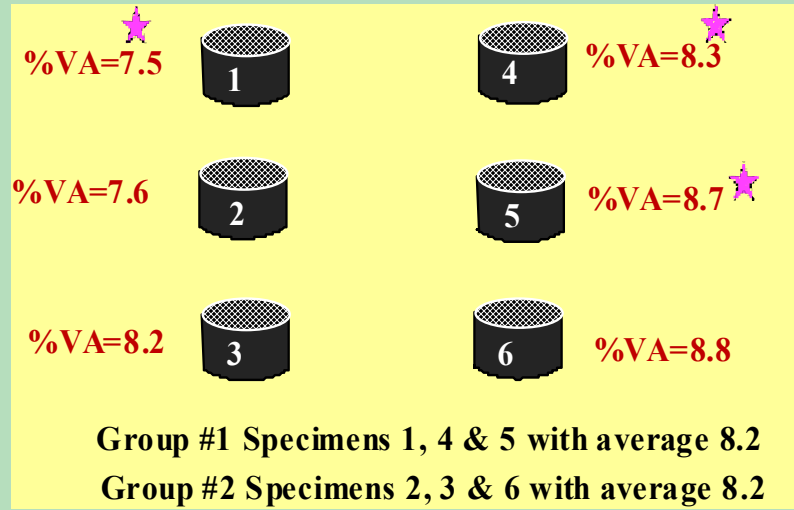


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Step # 8: Check Moisture Sensitivity (Optional)

- One set of the previously prepared two sets is used to determine the bulk specific gravity to make sure that the average air voids is 8.0 ± 1.0 . Partially saturate the first set in water at 25 °C in a vacuum desiccator for 10 minutes and measure the density to determine the percent saturation. Soak this set in thermostatically controlled water bath adjusted at 60 °C for 18 hours then transfer the specimens to another water bath adjusted at 25°C for two hours. During the two hours period, measure the density again and calculate the percent saturation and swell of the samples. The tensile strength of the specimens is determined by loading them in a loading jig until fracture occurs.
- The second set of specimens is stored in the laboratory and tested dry in the tensile loading jig. The tensile strength ratio of the two sets is calculated.

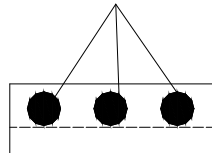
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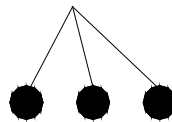
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Measuring Moisture Sensitivity

Average AV% (set#1) $\cong$ Average AV% (set#2)



Water Conditioned Specimens



Unconditioned Specimens

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Step # 9: Fatigue Testing (Optional)

- Fatigue behaviour of an asphalt layer depends not only on the properties of the mix but also on structural properties of underlying pavement layers, as well as traffic and climatic conditions.
- To perform the fatigue test, mix and condition sufficient material to make at least one slab 400x300x75 mm by using laboratory compactor. The air voids in the mix should be $5.0 \pm 0.5\%$. Using saw cut the slab to three beams each 400x63.5x50 mm as shown in Figure 5. Each of the triplicate beams is subjected to haversine loading at 10 Hz and $400 \mu\epsilon$. The failure is reached when the flexural stiffness drops to 50% of the initial value.

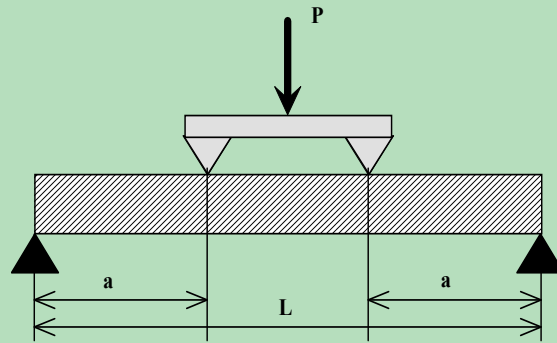
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Fatigue Test apparatus

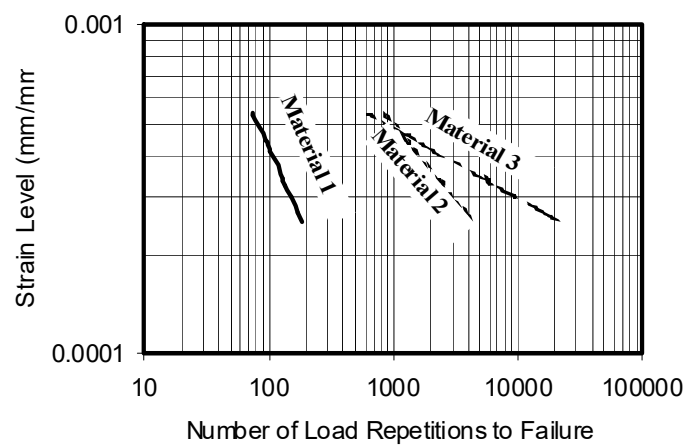


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Fatigue test principle

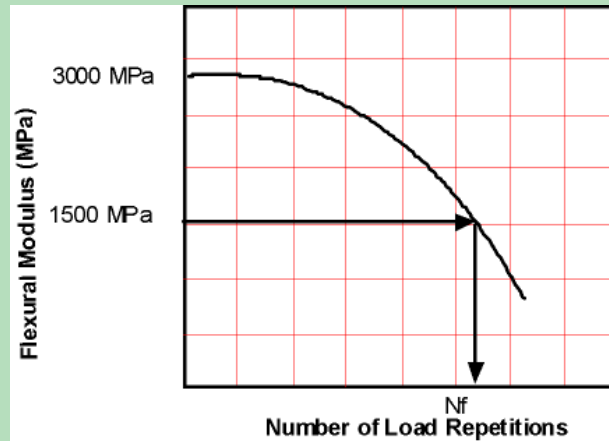


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Sample text slide



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Level 3

- In case of heavily trafficked situations, or where extra reliability in terms of rutting performance is required, level 3 testing is performed. This consists of two separate operations: a check of voids at 250 gyratory cycles, and the wheel tracking of slabs of the design mix.

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Voids check at the maximum cycles

- A high degree of compaction of a mix in the laboratory is an indicator of the final condition a mix may attain in the field if more severe conditions than expected are experienced (higher traffic levels or hotter pavement temperatures)
- Mixes with an air void content below 2.0 to 3.0% in service are likely to become unstable and prone to rutting.
- %VA @2 50 cycles $\geq$ 2.5%

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Wheel Tracking

- Sufficient material to make a 300 mm square slab, 50 mm deep (AC-14) or 75mm deep (AC-20) is mixed and conditioned in the normal manner. Two slabs will be required to obtain a single wheel tracking result.
- A laboratory compactor or footpath roller is used to compact the slabs. The air voids of the compacted slabs must be $5.0 \pm 0.5\%$.
- Wheel tracking test is carried out at 60°C using a wheel of 50 mm width and 200 mm diameter subjected to a load of 700 N.

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- Deformation rate is plotted against time and the slope of the line measured to determine the wheel tracking rate. If the wheel tracking rate of the two samples exceeds the specified performance criteria then the mix design process must be repeated using a more deformation resistant composition.

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